

HIRF Synthetic Environment: an innovative approach for HIRF design, analysis and certification of aircraft and rotorcraft

M. Bozzetti* (Alenia Aermacchi, Italy)
C. Girard (AxesSim, France)
A. Hoque (L-Up, France)
A. Marvin (York Univ., UK)

J.P. Moreoau (Dassault –Aviation, France)
J.P. Parmantier (ONERA, France)
M.S. Sarto (Sapienza Univ. of Rome, Italy)
C. G. Spiliotopoulos (Hellen Aerosp. Ind., Greece)

* contact author: mbozzetti@alenia.it

Abstract —HIRF SE (High Intensity Radiated Fields Synthetic Environment) is a computational tool for the simulation and analysis of electromagnetic field interaction with aircraft and rotorcraft in HIRF radio frequency range (10 kHz-40 GHz). The tool allows the simulation of air vehicle interaction with both far-field and near-field sources having any polarization and position in the presence or not of the ground, and it will be applied during design and certification phases in order to cope with the increasing complexity of air vehicles electronic/electric architectures and of the electromagnetic environment. The research leading to these results has received funding from the European Community's Seventh Framework Programme [FP7/2007-2013] under grant agreement no. 205294.

Keywords-component: HIRF, civil, aircraft, rotorcraft, design, certification

I. INTRODUCTION

Before air vehicles (aircrafts and rotorcrafts) can fly, International Certification Authorities (FAA for United States and EASA for Europe are the most influential worldwide) must be assured that:

- Equipment and subsystems are not affected by mutual electromagnetic interactions
- Subsystems/systems can correctly operate when subjected to external Electromagnetic disturbances.

The electromagnetic environment that air vehicle have to cope are defined in regulatory documents such as FAA 14 CFR Parts 23, 25, 27 and 29 and EASA Interim Policy and the ways to demonstrate air vehicle compliance are indicated in reference standards such as EUROCAE ED-107A/SAE ARP5583A "Guide To Certification of Aircraft in a High-Intensity Radiated Field (HIRF) Environment". It is common practice to demonstrate by testing air vehicle robustness against catastrophic effects induced by electromagnetic disturbances.

The current air vehicle lifecycle foresees the experimental verification at the end of the development phase when the air vehicles are already built and just before their release to the

market: this means that only during the certification phases it is possible to detect susceptibilities that correspond to weakness in the electromagnetic design. Rework costs are very high and even the time to market increases considerably once design needs to be review.

In the new generation air vehicles (fly by wire and even fly by wireless) the increasing number of function with catastrophic effect makes the time required for testing grow exponentially and in the meanwhile high speed electronic equipments poses serious problems to air vehicle safety verification due to EM hazard.

Moreover the electromagnetic environment is characterized by a proliferation of emitters that will to higher costs for instrumentation to carry out testing to comply with Certification Regulation demand.

In this scenario, numerical approach can help maintaining the duration and the costs of the testing within reasonable limits and can improve the accuracy of the measurements assessing the differences between free field illumination and ground test illumination.

HIRF SE is a Computational Framework with the ability to predict both the EM field distribution inside the airframe and the induced currents in the wiring system and will permit, by simulation, to:

- Anticipate air vehicles EM performances during the design phase
- Support certification testing verifying air vehicles robustness respect EM disturbances.

The benefic effects of HIRF SE applications during air vehicle lifecycle are a reduction of EM vulnerabilities, optimization of qualification EMC testing at equipment level and certification EMC test at air vehicle level and specific testing where the air vehicles needs to be tested (susceptibility analysis).



Fig. 1 HIRF testing inside Alenia Aermacchi Anechoic Chamber in Caselle Torinese premises

II. HIRF SE FRAMEWORK

HIRF SE framework will be an EM analysis tool operating in HIRF radio frequency range 10 kHz – 40 GHz. It will simulate the air vehicle interaction with both far-field and near-field sources having any polarization and position in the presence or not of the ground. Also HIRF SE will predict both the EM field distribution inside the airframe and the induced voltages and currents in the wiring system. The broadband EM perturbations will be simulated according to the following classification:

- External environment (i.e. radars, radio and television broadcast emitters).
- Internal environment (i.e. EM constraints in terms of EM fields and current on cables that air vehicle electronic installation will encounter due to external EM fields or internal emitter devices such as PEDs or WPEDs).

Three different EM sources can be distinguished and these are identified as threats:

- External EM threat (generated, for instance, by airport and radio-television emitters);
- Surface EM threat (generated by on-board emitters);
- Internal EM threat (generated by PED or WPED).

The aforementioned threats are the input of the transfer functions that HIRF SE will compute in the frequency range 10 kHz-40 GHz.

Simulation approaches have been classified with respect to a Low Frequency Band (LFB) scenario and a High Frequency Band (HFB) scenario.

- The LFB scenario extends from 10 kHz to 3 GHz. It relies on the use of 3D full wave solvers for calculating both external and internal environments of air vehicles and the use of Multiconductor Transmission Line models for the calculating the response of cable bundles under test. The process for linking the two types of data is based on the field-to-transmission line approach but this process still requires extensive thin wire modeling in the 3D model in order to describe correctly the inner field environment.
- The HFB scenario extends from 3GHz to 40 GHz. It relies on asymptotic 3D solvers for calculating external fields at

the level of points of entry of energy inside the air vehicle. The internal response is calculated with so called Power Balance tools making the assumption that the internal field environment can be considered as random.

In addition to those main computer modules, specific sub-modules have been developed in order to bring the EM physics required by HIRF modeling. Among them, material models have been extensively studied in order to be introduced as macro models in 3D computer solvers.

The HIRF SE framework uses the AMELET format in order to standardize the flux of data of the various computer models involved in the LFB and HFB scenarios.

The main output of HIRF SE will be the identification of possible susceptibilities (or the identification of frequency ranges without susceptibilities) following three steps:

1. Define the Environmental Level using HIRF SE to compute the EM environment caused by an external or on-board source and surrounding the equipment; The electromagnetic field around equipment is computed in the frequency range 100MHz-40GHz and the coupled currents flowing into equipment connectors are computed in the frequency range 10KHz-400MHz
2. Define the Susceptibility Margin comparing the computed environmental level with the equipment qualification level (100MHz-18GHz/40GHz) or system test level (10KHz-400MHz)
3. Assessment of Potential Susceptibility:
 - For all the frequencies in which the Susceptibility Margin is lower or equal to zero there is a potential susceptibility that need to be investigate by testing;
 - For all the frequencies in which the Susceptibility Margin is higher than zero it has been computed a minimum level (in term of current or EM field) of no susceptibility.

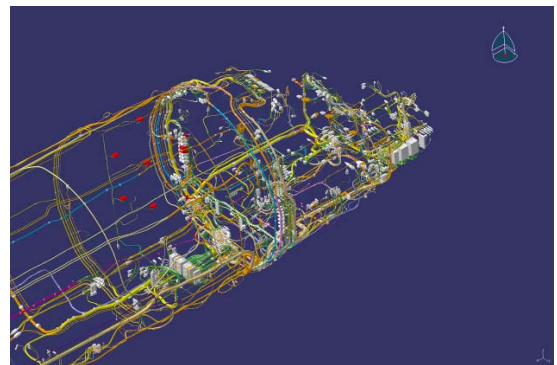


Fig. 2 Air vehicle model of cables, connectors and equipments

III. VALIDATION PROCESS

The correct operating of the framework will be verified and validated by comparison with data from real testing on small and medium air vehicles and pre-existing data for large air

vehicles. The validation at air vehicle level is the final step of the validation process.

Validation process is following a gradual approach in term of complexity in order to insure to have the control of the geometrical configuration while carrying out the validation of the tool that will be plugged inside the framework. In fact in the first phase of the validation the geometrical configurations of test objects are fully defined and controlled. The scope is to assess the validity of models, submodules and modules plugged into the HIRF SE framework without uncertainty regarding the test objects' geometry.

The test cases are classified in three different categories, corresponding to increasing complexity of the test object and each test case correspond to a HW test case that is used for generating experimental data to be compared with the results of the numerical simulation for assessment and validation of the relevant models submodules modules:

1. Test cases on "Generic Objects": they are finalized to assess the validity of the models developed for materials such as advanced composites, filled plastics, non structural materials (as fabrics or shielding protections) shielded cables and branch cables (e.g. Fig.3).

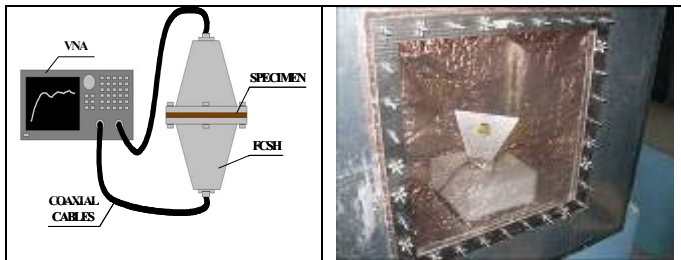


Fig. 3 Test Case on Generic Object: different testing set up for shielding effectiveness on expanded copper foil material.

2. Test cases on "Elementary Pieces": the test configurations for which the geometry and the EM characteristics are fully controlled. They are finalized to assess defined submodule of the framework. Examples of test cases are: bundles, equipment interfaces, nacelle structure, avionics bay-metal, empty fuselage, composite fuselage, structures made of composites and metal (e.g. Fig.4).

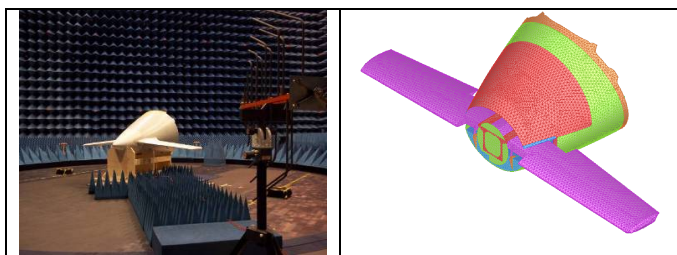


Fig. 4 Test Case on Elementary Piece: testing set up for shielding measurement and triangular meshed model of P180 nosecone

3. Test cases on "Sub-Systems": They are finalized to assess the HIRF SE modules when applied to the analysis of complex problems representing real configurations. Examples of test cases that have been considered are: complete equipment, instrumented equipment, avionics bay with cables, composite nacelle metal/filled, engine control system, avionics bay with cables and equipment-metal/composite, instrumental panel in metallic fuselage equipped with displays and bundles (Fig. 5).

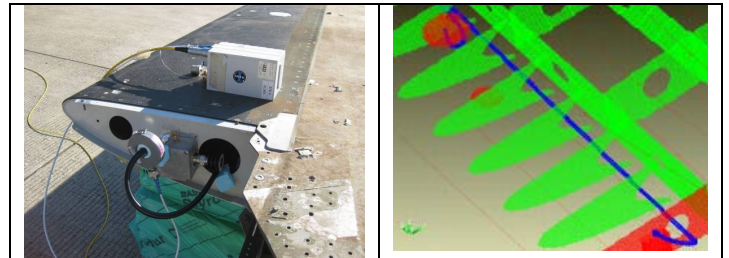


Fig. 5 Test Case on Sub-System: testing set up for coupled current and meshed model of Falcon F7X horizontal tail nosecone.

The validation of the framework at air vehicle level will be carried out with the goal to demonstrate that the solver is able to predict electromagnetic response of complex test objects in which the geometrical configuration is not completely under control.

The reference data for the comparison will be the measurement results, and in particular will be carried out along the project dedicated testing campaigns on small and medium air vehicles and in particular

- Medium air vehicles: Dassault F7X, EADS-CASA C-295;
- Small air vehicles: Piaggio P180 (Fig.6), PZL M28, Evektor EV55 and Cobra 100, Inta Siva;
- Rotorcraft: Agusta AW139, Eurocopter EC135, EC145.



Fig. 6 Test campaign on Piaggio P180 in Alenia Aermacchi's Anchoic Chamber and OATS

For the large air vehicles measurement campaign carried out before the starting of the project on BAE System Nimrod MRA4 will be used as references (Fig.7).



Fig. 7 Large air Vehicle: BAE System Nimrod MRA4.

IV. POSSIBLE APPLICATION DURING CERTIFICATION PROCESS

The common practice, as indicated by Advisory Circular AC-20158 and applicable standard ED107/A-July 2010, for certification of air vehicles can be summarized in three steps:

- Step 1: the air vehicle is exposed to an external Low Level HIRF environment.
- Step 2: evaluation of the internal environment around the equipment in terms of interface currents and in terms of electric field.
- Step 3: the internal environment values are compared with the equipment qualification/system integration results in order to verify compliance with HIRF certification requirements.

It is important to point out that

- The second step is normally achieved by means of measurements.
- The requirement for the air vehicles testing is for free field illumination (as close as possible).
- The test set-up may influence the illumination:
 - The plane-wave is a theoretical concept which does not exist in laboratory,
 - A ground plane is always in some vicinity of the air vehicle under test,
 - Laboratory illumination is not representative of the operative conditions.
- The amount of measurements typically achievable ranges between 100 and 1000. These figures might seem small in comparison of the actual complexity of critical systems.

Two different applications during air vehicle certification phases were presented and agreed with EASA: a short term

application and a long term application. The short term application will only use HIRF SE for tuning the test setup and compute corrections factors to apply to ground measurements to convert the data into free field environment data. These corrections factors will not only be based upon surface current but also on internal computed currents. The long term application will use direct computation of free field data that will be eventually calibrated by comparison between ground measurements and ground simulation.

The short term application:

- supports the Certification Process by determining the differences between free field set up and laboratory set up and determines internal A/C environment under free field condition
- keeps up with the increasing complexity without increasing the number of testing (frequency steps)

The long term application consists in:

- Reducing as much as possible testing in all the frequency ranges in which the adequacy of the results is demonstrated,
- Performing a limited number of measurements,
- Tuning models and verifying the reliability of modelled air vehicles

Consequently, in long term perspective, measurements will remain mandatory for calibrating the simulation tools but the majority of the testing for Certification purposes will be substituted by simulations.

V. CONCLUSION

With the development of HW architecture and of more efficient codes and algorithms, analysis and simulation are now affordable solutions for solving EMC problems in aeronautic and numerical modeling technique represent the most cost effectiveness approach for HIRF /EMC issues.

The aid of modelling and simulation will reduce the delivery timescales of future air vehicle and systems and simulation can embrace more complex situations than measurement raising to a higher level air vehicle safety.

In this context HIRF SE Framework is an innovative, collaborative and integrated methodology that allows solving EMC problem from CAD to analysis results in the frequency range up to 40 GHz and can be applied during early design and certification phases of air vehicle lifecycle.